Embracing Lag Uncertainty: Bayesian Model Averaging in Set-Identifed Structural VARs

Akın Aydın[*] Ferdinand Loser[†]

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Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

Keywords: Structural Vector Autoregression (SVAR), Set Identification, Bayesian Model Averaging (BMA), Model Uncertainty, Lag Order Selection, Monte Carlo Simulation
JEL Codes: C11, C15, C32, C52

[*]Ludwig-Maximilian University of Munich, akin.aydin@campus.lmu.de
[†]Ludwig-Maximilian University of Munich, ferdinand.loser@campus.lmu.de

1 Introduction

Introduce the problem of lag selection and parameter proliferation in Structural VARs. Highlight the debate between hard lag selection (Information Criteria) and Bayesian methods (Shrinkage and Model Averaging).

1.1 Bayesian Model Averaging (BMA)

We find it important to discuss the common properties of four different BMA approaches listed above. We follow the canonical treatment of Hoeting et al. (1999) for this part. First, we start with the problem of choosing the right weight parameters (so that we can average the models based on the weights). Let Δ be our parameter of interest (Impulse response functions) and let $M_1, \ldots, M_K$ be our candidate models (different reduced form VAR models). The posterior distribution of Δ given the data y (step 3 defined in the blueprint), averaged over all models, is:

$$p(\Delta \mid y) = \sum_{k=1}^{K} p(\Delta \mid M_k, y) p(M_k \mid y) \tag{1}$$

so this is the weighted average of the posterior distributions conditioned on the model $p(\Delta \mid M_k, y)$, weighted by $p(M_k \mid y)$ (posterior probability of model conditioned on data). By Bayes' theorem applied at the level of models, this weight is:

$$p(M_k \mid y) = \frac{p(y \mid M_k) p(M_k)}{\sum_{l=1}^{K} p(y \mid M_l) p(M_l)} \tag{2}$$

where $p(M_k)$ is the prior probability assigned to model M_k and $p(y \mid M_k)$ is its marginal likelihood—the likelihood integrated over the model's parameters under their prior:

$$p(y \mid M_k) = \int p(y \mid \theta_k, M_k) p(\theta_k \mid M_k) \, d\theta_k. \tag{3}$$

The marginal likelihood (or marginal data density) in equation (3) is central to our procedure, as it is the primary input to the model weights in (2).

These 3 equations defined above are the backbones of the Bayesian Model Averaging methodology. In practice, these three equations map directly onto the steps of our procedure. Equation (3)—the marginal likelihood—is evaluated first: for each candidate model we obtain a measure of how well it explains the data once its parameters are integrated out. For the BVAR models, indexed by a lag order and tightness pair (p, λ_1), this quantity is the marginal data density returned in closed form by the conjugate Normal-Wishart estimator; for the OLS models, indexed by lag order alone (p), it is approximated by the Bayesian Information Criterion and Akaike Information Criterion[1]. Equation (2) then converts these marginal likelihoods into model weights: each is multiplied by its model prior ($p(M_k)$)—uniform or geometric—and normalized so that the weights across all candidate models sum to one (denominator). These weights are the posterior model probabilities. Finally, equation (1) is realized not by analytical combination but by Monte Carlo: from each model we draw admissible impulse responses in a number proportional to its weight, so that models the data favor contribute more draws, and we collect all draws into a single pool. This pool is the Monte Carlo representation of the model-averaged distribution in (1). From it we select the median target as our single, coherent estimate of the impulse response, completing the averaging.

So far, we have described the common structure they share. However, we use four different BMA methods and each has its own priors or estimation methods. We discuss the differences below.

Weights and Priors

We use two weight sources and two model-space priors (uniform prior and geometric prior). Each combination yields a different metric:

	Uniform prior	Geometric prior ($\theta = 0.5$)
OLS (BIC/AIC-weighted, p)	Uniform-OLS	Geometric-OLS
BVAR (MDD-weighted, p, λ_1)	Joint Grid	Geometric Grid

We find it fruitful to discuss model priors that we use ($p(M_k)$) first. The uniform prior is straightforward. It represents the belief that every candidate model has the same probability of being the true model (reflecting the absence of any prior preference among lag orders):

$$p(M_1) = p(M_2) = \cdots = p(M_K) = \frac{1}{\bar{p}} \tag{4}$$

On the other hand, geometric prior constructs the prior as:

$$p(M_k) \propto \theta^p \quad \text{where} \quad \theta = 0.5 \tag{5}$$

We set $\theta = 0.5$, under which each additional lag halves the prior weight—equivalently, the prior demands that the marginal likelihood favor a longer model by at least a factor of two before the additional lag is preferred. This value is a deliberate, interpretable benchmark rather than an optimized choice: it imposes a moderate preference for parsi-

[1] Steel (2020,§3.7) mentions that marginal likelihood is approximated by Bayesian Information Criterion when closed form of equation (3) does not exist as BIC asymptotically converges to marginal likelihood. Also, Burnham & Anderson (2002) justifies the usage of AIC as an approximation of Marginal Likelihood. Therefore, we use both criteria.

mony, neither so weak as to coincide with the uniform prior nor so strong as to dominate the evidence from the marginal likelihood.

We use BIC and AIC for the models estimated via OLS. The reason is that it is not possible to calculate marginal likelihood since we don't have any priors (we don't form any beliefs regarding the parameters (no shrinking) as we use exactly one degree of freedom for each parameter we estimate under frequentist approach). We therefore approximate it by the Schwarz criterion, which Schwarz (1978), Raftery (1995) derived as an asymptotic approximation to the log marginal likelihood and which, in large samples, does not depend on a prior. For the BVAR models, by contrast, the Minnesota prior (discussed in paper A) supplies exactly the density (3) requires, and the conjugate structure makes the integral available in closed form, so the marginal data density is computed directly.

Advantages of the Bayesian Model Averaging

First of all, Bayesian Model Averaging is more honest about the model uncertainty. Following Hoeting (1999):

$$\mathrm{Var}[\Delta \mid y] = \sum_k \left(\mathrm{Var}[\Delta \mid y, M_k] + \hat{\Delta}_k^2\right) p(M_k \mid y) - E[\Delta \mid y]^2 \tag{6}$$

Here, $\hat{\Delta}_k = E[\Delta \mid y, M_k]$. After decomposing $\mathrm{Var}[\Delta \mid y]$, it is possible to see that BMA framework accounts for the model uncertainty:

$$\mathrm{Var}[\Delta \mid y] = \underbrace{\sum_k \mathrm{Var}[\Delta \mid y, M_k] p(M_k \mid y)}_{\text{within-model}} + \underbrace{\sum_k \left(\hat{\Delta}_k - E[\Delta \mid y]\right)^2 p(M_k \mid y)}_{\text{across-model}} \tag{7}$$

We can see from (7) that the perfect model satisfies $\hat{\Delta}_k = E[\Delta \mid y]$ so $\mathrm{Var}[\Delta \mid y]$ is only equal to within-model variance. That is clearly an advantage of Bayesian Model Averaging since it allows the practitioner to estimate the posterior variance in a more accurate way.

Second, its predictive ability is better compared to conventional methods (even though our work doesn't contain anything regarding forecasting, it's a strength of the methodology that we followed). Madigan-Raftery (1994), claims that under log-scoring BMA's predictive ability is better. Also Hoeting (1999) demonstrates a concrete example and shows that even the best performing model has a 17% posterior probability weight. Therefore, it makes sense to use Bayesian Model Averaging in order to account for model uncertainty. Our main motivation to use this approach is to see whether accounting for model uncertainty leads to better MSE results or not.

2 Simulation Design

This Monte Carlo simulation study follows Ivanov and Kilian (2005) design with several alterations in order to make our extensions feasible and to fit the needs of our analysis. First of all, we only use one monthly DGP based on the empirical VAR study of Kilian and Murphy (2014). Therefore, we do not get estimations for quarterly VAR series or VEC series, and additionally, we do not have additional robustness by including several different series with different macroeconomic indicators and movements. The main reason for this is our tight time constraint. However, this concern is also not to be overstated as Ivanov and Kilian (2005) did not find big differences between different DGPs of the same type (monthly, quarterly, VEC). We use the dataset from Kilian and Murphy (2014) containing the following 4 variables: percent change in global oil production, real activity index from Kilian (2009), the log real price of oil, and changes in OECD crude oil inventories. Just as in the original study, we first estimate a VAR with a set lag order of these 4 variables and then use the sign restrictions approach explained earlier to identify structural IRFs. Therefore, our estimation becomes much more complex and updates the findings of Ivanov and Kilian (2005) to a more modern and widely used approach in the literature. All papers used for the DGP generation in Ivanov and Kilian (2005) are from the 1990s and early 2000s and are based on the Cholesky decomposition approach for identification. Cholesky decomposition only gives one unique set of IRFs, whereas the sign restrictions approach principally has an infinite number of possible sets of IRFs.

We follow Kilian and Murphy (2014) in searching for a large number of admissible IRFs by randomly drawing from the space of rotation matrices. We use exactly the same sign restriction matrix, but limit the number of periods for which dynamic sign restrictions are imposed to 3 instead of 12, and do not include additional elasticity bounds. This is a significant diversion from the original study as it makes the identification more agnostic and opens up the space of admissible IRFs. We are therefore closer to the original sign restrictions approach of Uhlig (2005). We understand that the additional restrictions of Kilian and Murphy (2014) are motivated by limiting the set to fit theory and other empirical evidence, we have to take the less restrictive approach in order to make the estimation feasible. Especially for low lag orders and smaller samples, these restrictions have proven to be so strong that even several million draws of the rotation matrix do not give us enough admissible IRFs to make a good inference. For the simulation, this exceedingly high number of draws would need to then be repeated hundreds of times.

Out of this set, we select the median target IRF as our true IRF, which is defined as the IRF that has the smallest distance to the median of all admissible IRFs. The DGP follows this IRF as it is generated by a structural VAR model using the structural impact matrix $\tilde{B}$ that corresponds to the median target IRF as the parameter for the structural shocks e_t, which is drawn from an i.i.d. standard normal distribution. The formula for the DGP is represented by a structural VAR model using the estimated reduced form parameters A (for the lag coefficients y_{t-p}) and V (for the monthly dummy parameters d_t) and the structural impact matrix $\tilde{B}$ as follows:

$$y_t = Vd_t + Ay_{t-p} + \tilde{B}e_t \tag{8}$$

We estimate these parameters using the original data for five different lag orders, $p_0 \in 4, 6, 8, 10$ and create DGPs with different sample sizes, $T \in 240, 300, 360, 480, 600$, where $T = N + \bar{p}$. $\bar{p}$ is the set of lag orders that are considered for both the IC and the BVAR estimations; it is uniformly set to $\bar{p} = 12$ to ensure that the true lag order p_0 is part of the set. These specifications directly follow Ivanov and Kilian (2005). We do not increase the number of lags or higher sample sizes in order to keep the comparison intact and to keep computational capacities available for the switch to sign restriction decomposition and the addition of BVAR. For sampling, we use random draws of length p_0 from the original dataset as initial values, then apply the parametric bootstrap using the SVAR described above and generate sequences of lengths T.

On each of these samples, OLS VARs fitted by lag orders selected by both the three ICs as well as the BMA determined OLS VARs and BVARs, for both the geometric and BIC-based approach, are estimated, a fixed number of admissible IRFs found, and the median target IRF determined. We only include AIC, BIC and HQC as information criteria and omit the Likelihood Ratio tests and the Lagrange Multiplier test for lag selection as these have become practically irrelevant in the last 20 years for macroeconomic time series analysis and were also shown to perform poorly in Ivanov and Kilian (2005). The four different BMA estimations follow the methodology described in the previous section. The space of optimal shrinkage parameters for the implementation with BVAR is limited to $\lambda_1 \in \{0.05, 0.2, 0.35, 0.5, 0.65, 0.8, 0.95\}$ in order to make computation in the simulation feasible. It still captures a good range of typically used $lambda_1$-values, both tighter and looser specifications.

Finally, the squared error matrix of the selected median target IRF to the true DGP IRF is calculated, collapsed into a scalar total squared error by summation. Then we estimate the ratio of this squared error of a specific estimation approach to the model fitted by the true lag order p_0. Therefore, the deviations of the true IRF are directly put into relation to how well a correctly specified model compares for the given sample. Finally, we estimate the MSE by calculating the geometric mean of these individual ratios. The geometric mean is preferred over the arithmetic mean as the ratios are bounded on the left side at 0, but unbounded on the right side. Outliers on the right side are hence overweighed in the arithmetic mean. The relative MSE for one method for one true lag order p_0 and one sample size T is given by the following formula:

$$MSE_{relative} = \exp\left(\frac{1}{n}\sum_{i=1}^{n}\ln(r_i)\right) \quad \text{where } r_i = \frac{TSE_i}{TSE_i^{p_0}} \tag{9}$$

While, for a single estimation it is possible to estimate millions of admissible IRFs, the Monte Carlo simulation approach doesn't allow for such a large number of draws. In establishing the median target IRF specific to the DGP, we set the number of discovered admissible IRFs to 1 million so as to have a very accurate median target IRF for the specific dataset. In the Monte Carlo Simulation, we reduce this number drastically to 100 as it has proven to be the absolute limit that still allows us to have an acceptable number of Monte Carlo iterations, 1000, for our set of different methods. Even with computational optimization by parallel processing, more draws are not applicable under our hardware constraints. For each individual iteration, the median target IRF is therefore less accurately determined. However, because we use relative MSEs as our performance measure and repeat the process 1000 times, this should not impact the interpretability of our results.

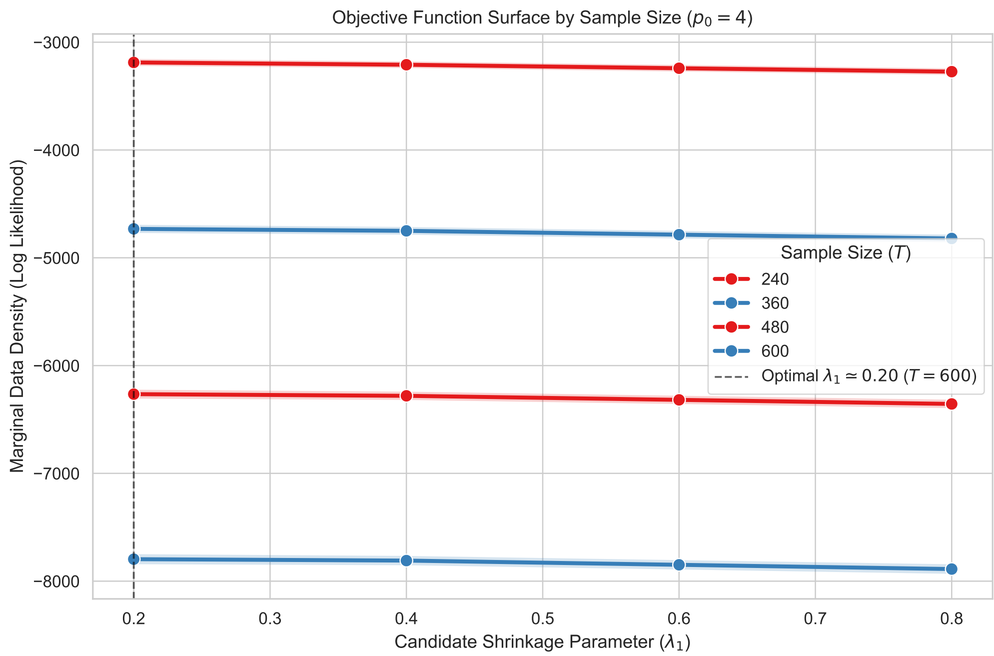


Figure 1: Objective Function Surface for Marginal Data Density Optimization. The peak represents the optimal balance between model fit and parameter penalty.

3 Results

First, we will look at the main comparison of performance between the different ICs and BVAR specifications in terms of recovering an accurate median target IRF. Table 2 displays the full set of relative MSEs. It reveals two main findings:

1. We confirm the results from Ivanov and Kilian (2005) about the performance of the different information criteria. AIC consistently outperforms both BIC and HQC, revealing that the strict penalization of the latter two towards low lag orders is biasing the IRF too much. On the other hand, AIC's milder lag penalization ensures a closer fit to the true lag order p_0 for finite samples, thereby reducing bias. The reduction of the variance by selecting a p much lower than the true lag order selected by BIC and HQC, see Table 1, is not evidenced to reduce the coefficient variance enough to compensate for the additional bias.

2. Hedging the risk of lag selection by employing BMA is the best option in nearly all scenarios. For smaller sample sizes, $T = 240$ and $T = 360$, joint grid BMA is dominating all other methods by a solid margin. The OLS fitted BMA using AIC as criterion is the best criterion in two scenarios, once without and once with geometric decay. However, for $p_0 = 10$ and $T = 600$, joint grid BMA's relative MSE is only slightly higher. Only for the highest sample size $T = 600$ and the lower true lag order $p_0 = 4$ does the conventional VAR fitted by an AIC-selected lag order perform the best. This may be explained by the finding that $\hat{p}^{AIC}$ has basically converged to p_0 at this sample size (See Table 1) while the sample size is already large enough in order to accurately determine the coefficients.

Table 3 makes the comparison to AIC for all different scenarios clearer by setting the MSE_{AIC} as the benchmark. Both OLS BMA with AIC as well as joint grid BMA outperform regular AIC in the majority of scenarios. All other versions of BMA are actually not consistently better than simple AIC. How can it be that the advantage of BMA hedging depends so heavily on the exact specification?

To understand this, we have to consider Figures 3 and 4 that present the weighting distribution of the different BMA methods for $p_0 = 4$. The reason for BIC-weighted OLS BMA to fail is very apparent. The strong penalty of BIC forces it to rely very heavily on the lag order $p = 2$ in finite samples, thereby basically eliminating the diversification of BMA. As BIC generally doesn't produce models with good IRF inference, this thereby also applies to these BMAs.

Once you introduce AIC to OLS BMA, the goal of spreading out the estimation on a selection of lag orders that are determined to be probable is achieved. AIC's weaker penalization of higher lag orders ensures that four to five different ps are assigned positive weights. The introduction of the geometric decay parameter $\theta = 0.5$ distorts the weighting towards lower lag orders. This moves the estimation further away from the correctly fitted model resulting in a larger underfitting bias. Table 3 shows that the reduction in parameter variance does not compensate this bias.

Finally, the joint grid approach provides the most widely distributed weighting scheme. For lower sample sizes, nearly all lag orders are considered. And even for the largest $T = 600$, the highest weight is just above 50 percent. Hence a wider diversification, also into the territory of overfitting lag orders $p > p_0$, can improve the fit of the determined median target IRF if combined with BVAR estimation.

3.1 The Uncertainty Strategy: Averaging vs. Selection

Here, we evaluate True Bayesian Model Averaging (BMA) against Information Criteria. We focus on how BMA acts as an insurance policy against catastrophic lag misspecification in small samples.

Figure 3 reveal how the weights are spread out for the four different BMA methods. Both OLS-based approaches produce a much more concentrated weighting scheme as higher lag orders are completely eliminated due to their high parameter variance in OLS-estimation. High lag orders are found to still introduce too much variance into the estimation to suffice a non-zero weight. On the other hand, both BVAR-based approaches spread out the weights much wider, and do not set high lag orders to 0. The additional parameter shrinkage of the BVAR estimation therefore enables BMA to still include more lag orders as their parameters are further shrunk towards 0. This shrinkage is further adapted to the specific chosen lag order p as both geometric and joint grid BMA jointly optimize p and the p-specific λ_1-value. A higher lag-order can therefore be combined with a tighter λ_1-value in order to shrink down the higher lags more drastically.

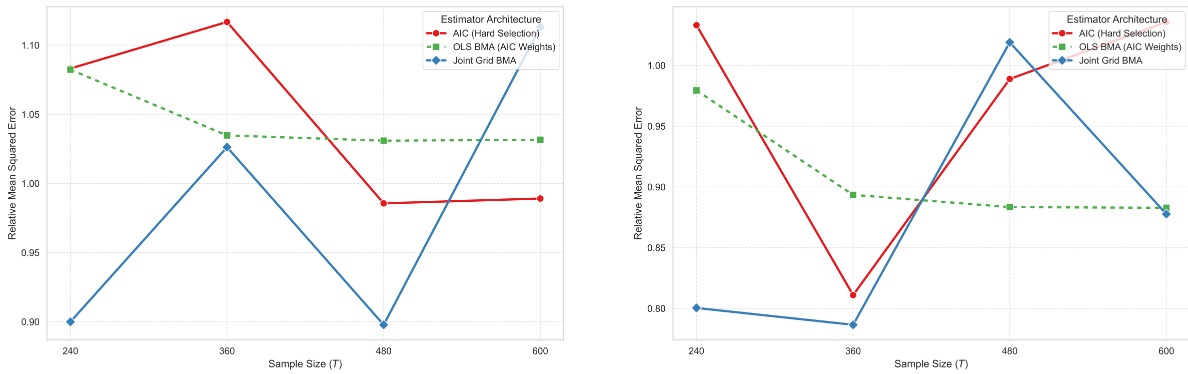
5 illustrates how smaller λ_1-values are selected for higher lag orders p.

Table 1: Relative Mean Squared Error: Hard Selection vs. Bayesian Model Averaging (Including Informative Priors).

T	p_0	AIC	BIC	HQC	OLS BMA (BIC)	OLS BMA (Geom. BIC)	OLS BMA (AIC)	OLS BMA (Geom. AIC)	Joint Grid BMA	Geom. Grid BMA
240	4	1.083	1.108	1.129	1.100	1.163	1.082	1.175	**0.900**	1.127
	10	1.033	1.018	1.021	1.010	0.842	0.979	0.982	**0.800**	0.849
360	4	1.117	1.274	1.296	1.142	1.235	1.035	1.138	**1.026**	1.046
	10	0.811	0.904	0.878	1.163	1.057	0.893	0.884	**0.787**	0.993
480	4	0.986	1.133	1.124	1.148	1.150	1.031	0.996	**0.898**	1.130
	10	0.989	1.190	1.185	1.052	1.283	**0.883**	1.210	1.019	0.898
600	4	**0.989**	1.145	1.118	1.218	1.129	1.032	1.124	1.113	1.109
	10	1.036	1.260	1.259	1.293	1.100	0.883	**0.875**	0.878	0.964

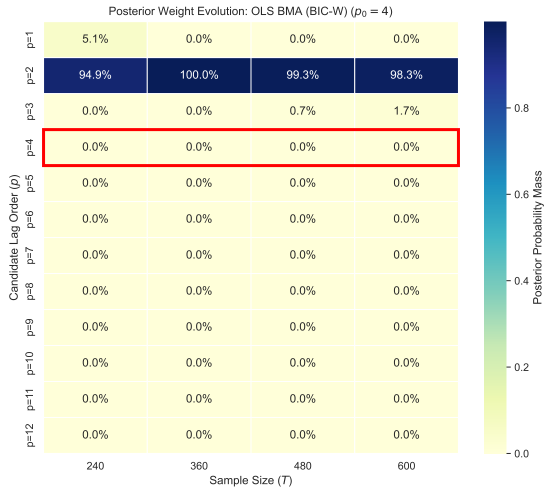
Table 2: Relative Mean Squared Error of Bayesian Model Averaging Specifications vs. AIC Baseline.

T	p_0	OLS BMA (BIC)	OLS BMA (Geom. BIC)	OLS BMA (AIC)	OLS BMA (Geom. AIC)	Joint Grid BMA	Geom. Grid B
240	4	1.015	1.074	0.999	1.085	**0.831**	1.040
	10	0.978	0.815	0.948	0.950	**0.775**	0.821
360	4	1.023	1.106	0.927	1.019	**0.919**	0.937
	10	1.434	1.304	1.102	1.090	**0.970**	1.224
480	4	1.164	1.166	1.046	1.010	**0.911**	1.147
	10	1.064	1.297	**0.893**	1.224	1.031	0.908
600	4	1.231	1.141	**1.043**	1.136	1.125	1.122
	10	1.248	1.062	0.852	**0.845**	0.847	0.930

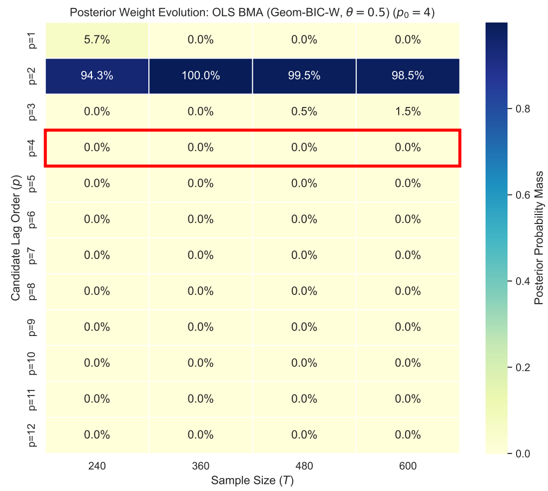


(a) $p_0 = 4$ (b) $p_0 = 10$

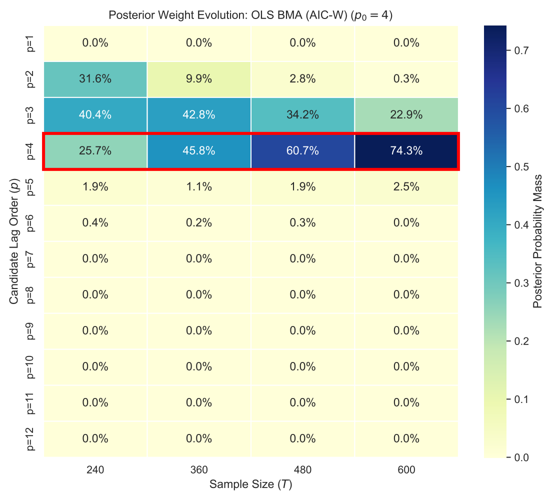
Figure 2: Performance of BMA vs Hard Selection at different sample sizes T.



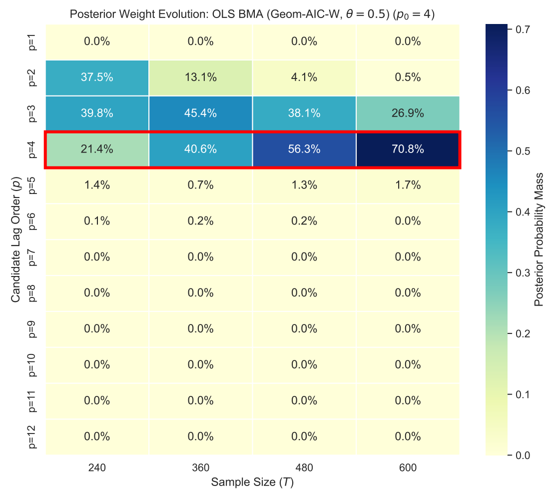
(a) OLS BMA (BIC-weighted)



(b) OLS BMA (Geom. BIC-weighted, $\theta = 0.5$)

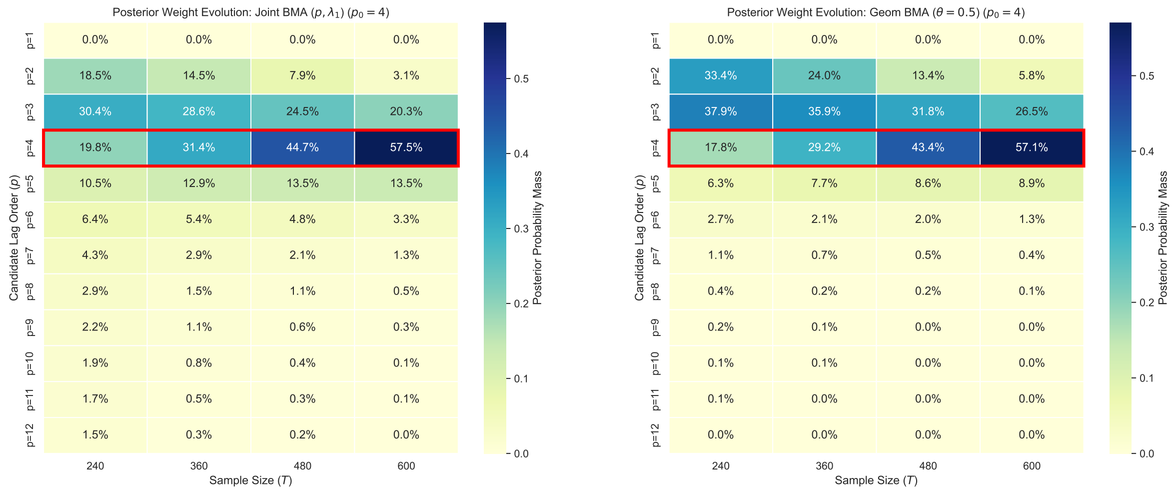


(c) OLS BMA (AIC-weighted)



(d) OLS BMA (Geom. AIC-weighted, $\theta = 0.5$)

Figure 3: Posterior Weight Evolution across the four OLS BMA architectures ($p_0 = 4$).



(a) Joint Grid BMA (b) Geometric Grid BMA

Figure 4: Posterior Weight Evolution for the two BVAR-based BMAs ($p_0 = 4$).

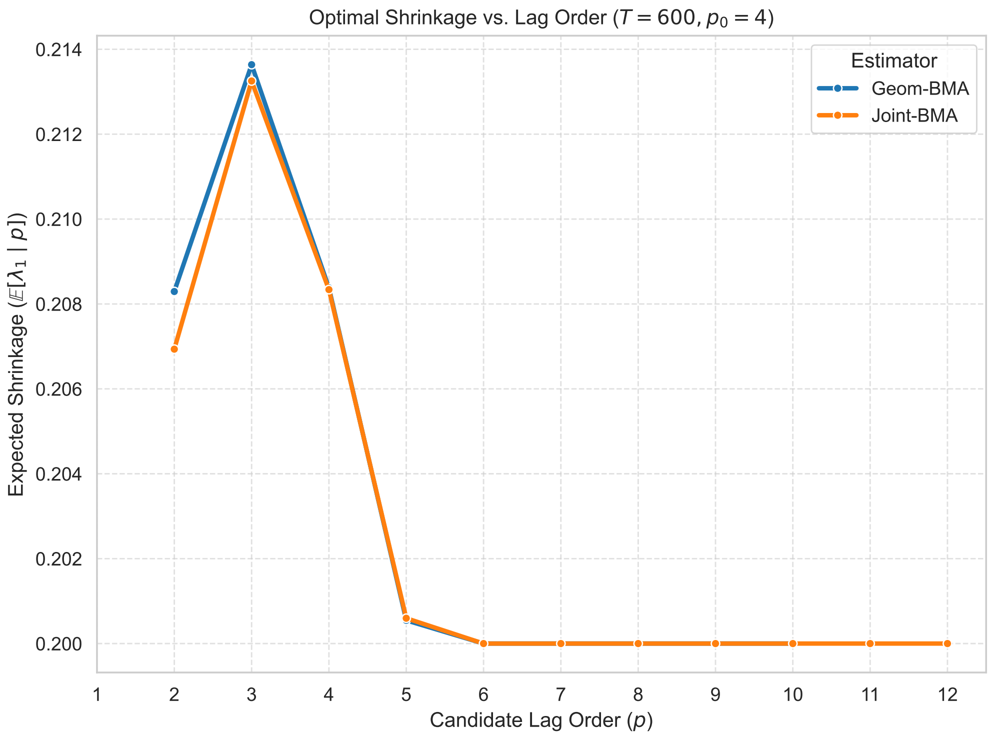


Figure 5: Optimal λ_1 is adapted to the lag order p.

4 Conclusion

Summarize the main findings, specifically highlighting how optimal τ selection and true BMA systematically outperform standard econometric heuristics in finite samples.

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Declaration of AI Usage

During the preparation of this seminar paper, I, [NAME], used Claude, Gemini, and ChatGPT for help with coding, research, and adapting writing to academic standards. I have reviewed and edited the content as needed and take full responsibility for the content.

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